

COM 837: ML for Wireless Communication Systems

Assignment 2 Report

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GitHub Repository: https://github.com/Brijesh-Coder-IIITB/MLWC_Assignment2

Question 1 - SVM for LOS/NLOS Identification

Part (b) Observations

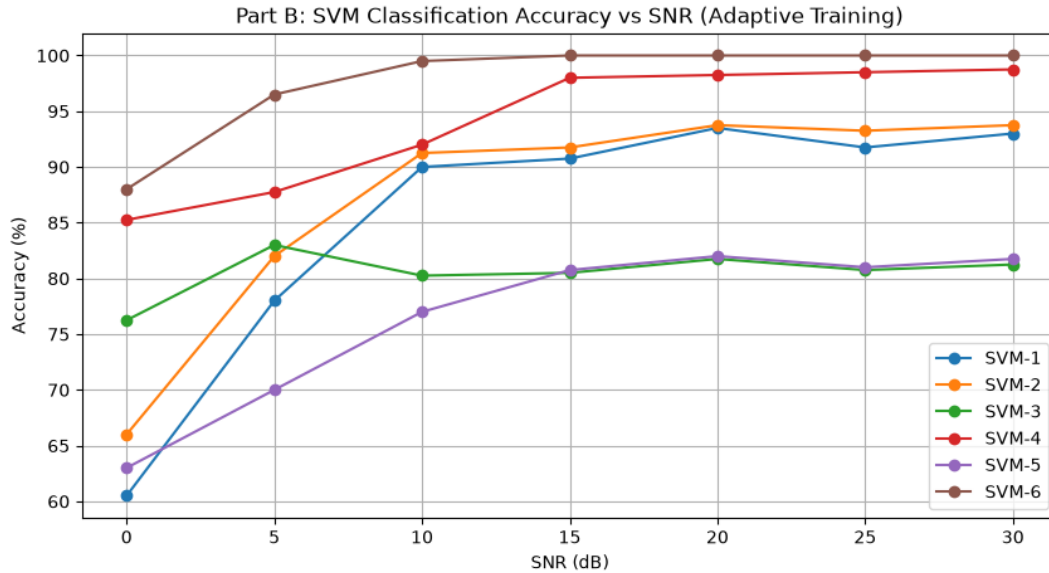


Figure 1: SVM Classification Accuracy vs SNR (Adaptive Training)

- **Figure 1 observation:** The Root Mean Square (RMS) delay spread (SVM-4) emerges as the strongest single feature overall. It reliably captures the physical difference between the direct LOS path and the dispersed time delays characteristic of heavily scattered NLOS environments.

The combined five-feature model (SVM-6) shows noticeable improvement over the best single feature, particularly in the 0 to 15 dB range. This demonstrates the value of combining complementary channel features to resolve complex decision boundaries.

- **Low-SNR performance:** The Rician K-factor (SVM-5) is the lowest-performing feature at severe noise levels (0 dB). Heavy thermal noise introduces random amplitude spikes that the mathematical formula misinterprets as a dominant LOS path, making the statistic unreliable in high-noise conditions.

Part (c) Observations

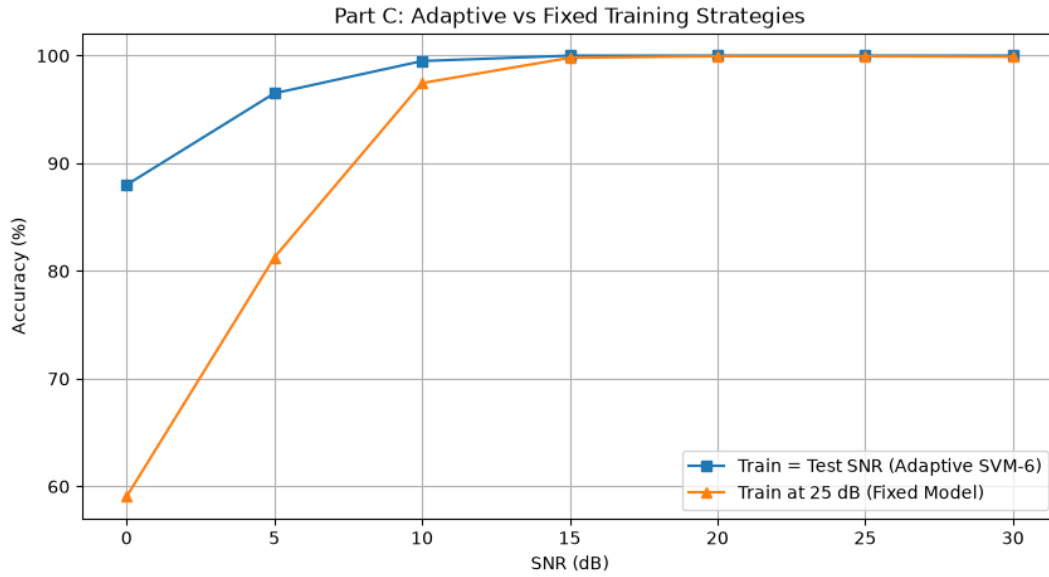


Figure 2: Adaptive vs Fixed Training Strategies

- **Figure 2 observation:** The matched-SNR (adaptive) model performs better at low SNR because it learns feature distributions that include the severe distortion present at those operating conditions.

The model trained exclusively at 25 dB learns boundaries from cleaner features, causing it to lose accuracy when applied to heavily noisy 0–10 dB data. At high SNRs (20–30 dB), the two curves converge because the feature distributions stabilize as the noise becomes small.

- **Low-SNR interpretation:** Training exclusively at 0 dB and testing across all SNRs would create a severe distribution mismatch. The model would learn boundaries heavily influenced by noise variance and then misapply them to tightly clustered, clean features at high SNRs.

Question 2 - Constellation Demodulation & Clustering using K-means

Part (b) Observations

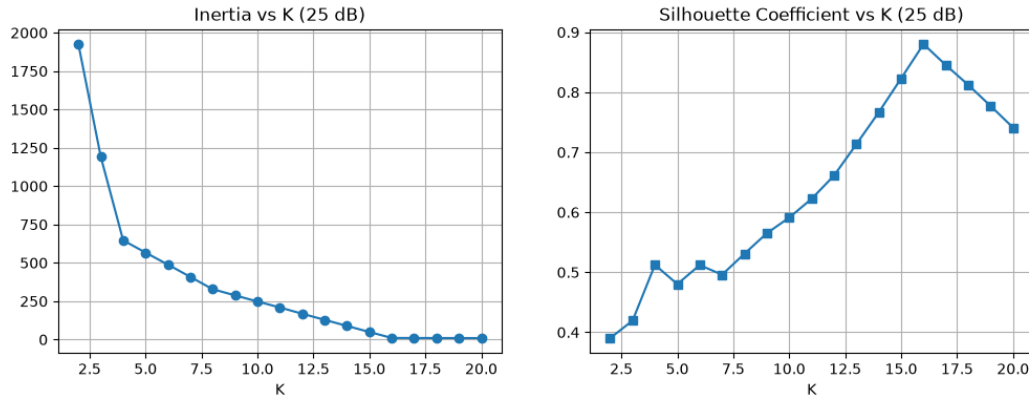


Figure 3: Inertia and Silhouette Coefficient vs K (25 dB)

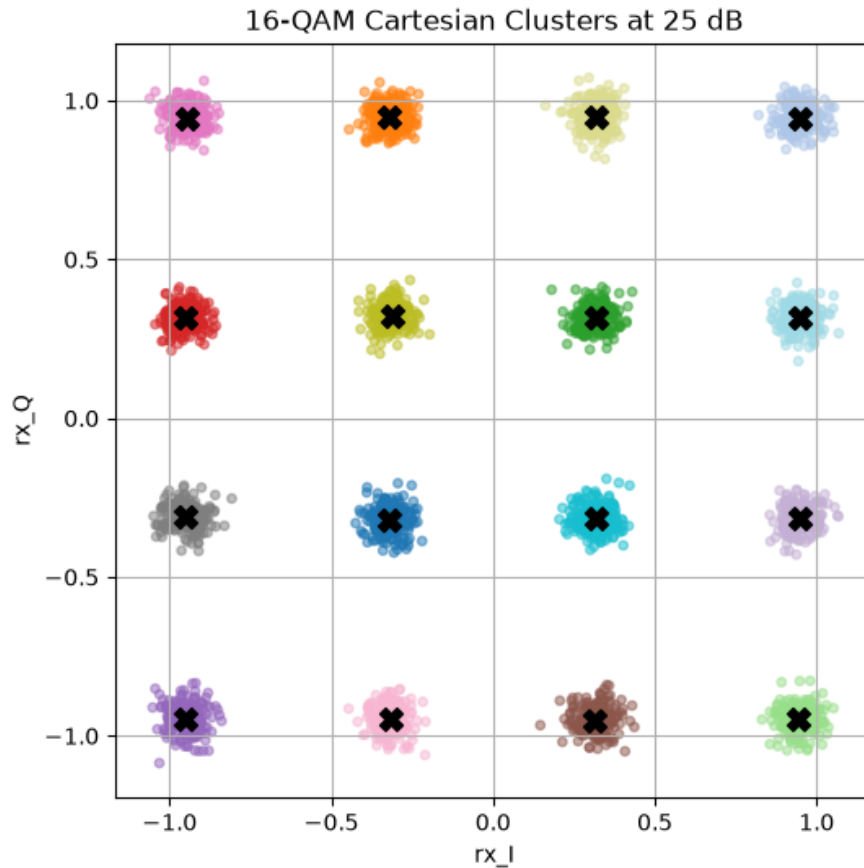


Figure 4: 16-QAM Cartesian Clusters at 25 dB

- **Figure 3 observation:** Inertia decreases as K increases, with the main reduction occurring before the true 16-symbol structure is represented. The silhouette curve confirms this cluster separation, validating $K = 16$ as the appropriate choice based on both diagnostics and the known constellation size.

- **Figure 4 observation:** At 25 dB, the received samples form 16 compact groups around the QAM grid, and the learned centroids accurately align with these groups.
- **Geometry observation:** Feature Set 1 (Cartesian coordinates) produces the highest Cluster Purity because Cartesian Euclidean distance perfectly matches the physical grid geometry of square QAM. Conversely, unweighted polar coordinates yield lower purity because equal changes in angle do not represent equal Cartesian distances at different radii, warping the natural decision boundaries.

Part (c) Observations

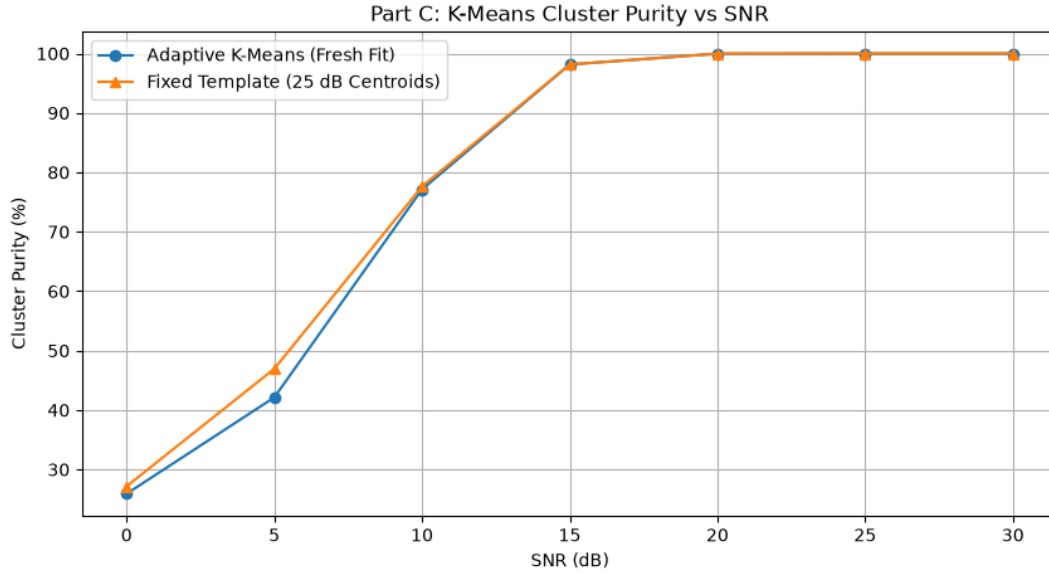


Figure 5: K-Means Cluster Purity vs SNR

- **Figure 5 observation:** At high SNR, both the adaptive K-means and fixed template strategies approach identical purity levels because the received points remain tightly clustered close to the ideal constellation.
- **Low-SNR behavior:** At low SNRs (0–10 dB), neighboring symbol clouds heavily overlap. This causes adaptive centroids to shift or merge toward artificially dense noise regions, abandoning the true grid layout.
- **Template stability:** The fixed 25 dB centroid strategy achieves higher classification accuracy at low SNRs because it rigidly retains the known geometric structure of the grid, making it more stable against severe noise overlap.